

Soundscape-Validated Ensemble Selection for Bioacoustic Species Identification in the Pantanal Wetlands

Kanta Saito

Dept. of Computer Science

Univ. of Hawai‘i at Mānoa

kantas@hawaii.edu

Abstract

We present our entry to the BirdCLEF+ 2026 challenge, a multi-label acoustic species identification task on continuous Pantanal soundscape recordings. Our system combines historical pretraining, soundscape-domain background augmentation, weight averaging across cross-validation folds, and an ensemble of three ImageNet-pretrained CNN backbones evaluated under two log-mel configurations. It reaches a public-leaderboard macro-averaged ROC-AUC of 0.833, up from 0.705 without soundscape augmentation. Our central contribution is a selection procedure. We observe that out-of-fold (OOF) AUC on held-out Xeno-Canto clips is anti-correlated with leaderboard AUC. Models the OOF tuner upweighted score below 0.65 macro-AUC on 66 labeled Pantanal soundscape files, while models it downweighted score above 0.96 on the same audio. We propose soundscape-validated ensemble selection, which tunes ensemble weights on the labeled soundscape distribution rather than on OOF predictions, yielding a +0.018 leaderboard improvement in a single submission. Adding soundscape AUC as the early-stopping signal during training contributes a further +0.003 leaderboard AUC. We also document two negative results: pseudo-labelling on unlabeled soundscapes consistently reduced LB AUC by 0.07 to 0.09, and per-class ensemble weights produced a fit-set gain that collapsed to noise under leave-half-out cross-validation across the 66 labeled files. Code is available at <https://github.com/KantaS12/birdCLEF2026>.

1. Introduction

The BirdCLEF series [1, 2] is the standard benchmark for large-scale bioacoustic species identification. The 2026 edition focuses on the Pantanal, the world’s largest tropical wetland and one of its most biodiverse regions. The task is to predict, for each 5s window of a 1 min field recording, the presence of any of 206 wildlife species spanning birds, mammals, amphibians, insects, and reptiles. Performance is measured by macro-averaged ROC-AUC, so a species with a single training clip contributes equally to the metric as one with 499 clips. Inference must complete on a CPU-only Kaggle kernel within a 90 min budget for the entire hidden test set.

The dominant difficulty in this competition is a domain gap. Training audio is drawn predominantly from Xeno-Canto [3], which provides short, single-species, volunteer-recorded clips in quiet conditions. Test audio consists of long, unsegmented soundscapes recorded by autonomous acoustic recorders deployed in Pantanal sites. The two distributions overlap weakly. The test side contains overlapping calls, rain, wind, insect noise, and microphone characteristics absent from training. A second difficulty is the long-tailed label distribution. Per-species clip counts span three orders of magnitude (1 to 499 clips per species, median 125), and training without rebalancing collapses on the rare classes that the macro-averaged metric is sensitive to. A third, less obvious difficulty is that cross-validation folds drawn from the training distribution do not match the deployment distribution. Selecting ensembles on out-of-fold AUC actively promotes the models that generalize worst (Section 4.3.).

Adopted versus new. Our backbone choices (EfficientNetV2-S, `eca_nfnet_l0`, EfficientNet-B0) [4–6], the SED head with attention pooling, log-mel spectrogram inputs at 32 kHz, focal-BCE loss with label smoothing [7], mixup [8], SpecAugment [9], cross-fold weight averaging [10], and overlapping-window inference are adopted from recent BirdCLEF winning solutions and the `timm` [11] ecosystem. The new components introduced in this work are (i) soundscape-validated ensemble selection (Section 3.3.), (ii) soundscape-aware early stopping during training (Section 3.4.), and (iii) the empirical characterization of the OOF and leaderboard anti-correlation (Section 4.3.) along with two clean negative results that follow from the same mechanism.

2. Data

2.1. Dataset

We use the BirdCLEF+ 2026 training set, which contains 35,549 audio clips covering 206 species (34,799 Aves, 451 Amphibia, 199 Insecta, 99 Mammalia, and 1 Reptilia). The distribution is long-tailed, with median 125 clips per species and a range of 1 to 499. Clips are short, under 30 seconds, and consist of single-species recordings drawn predominantly from Xeno-Canto [3].

The competition additionally provides 66 labeled one-

minute soundscape files recorded by autonomous acoustic recorders in the Pantanal, together with approximately 10,500 unlabeled soundscape files from the same recorder fleet that we use only as a source of background audio for augmentation. The hidden test set consists of around 700 unlabeled one-minute soundscapes from the same fleet.

2.2. The Triple Role of the Labeled Soundscapes

We use the 66 labeled soundscapes in three roles, kept deliberately distinct in the discussion below.

- **Training samples.** Concatenated to the labeled clip pool through a separate Hierarchical Data Format version 5 (HDF5) input.
- **Early-stopping signal** during fine-tuning (Section 3.4.).
- **Validation distribution** for ensemble weight selection (Section 3.3.).

Because all three roles draw on the same 66 files, the soundscape AUC numbers we report are an upper bound on out-of-distribution generalization rather than a held-out estimate. The public leaderboard remains the only fully held-out signal.

2.3. Audio Preprocessing

All audio is resampled to 32 kHz. Clips shorter than 5 s are zero-padded, and longer clips are randomly cropped during training and centre-cropped at validation. We use two log-mel configurations and train each backbone twice, once per configuration, for ensemble diversity at inference.

- **Config A.** $n_{\text{mel}}=128$, $\text{hop} = 320$, $f_{\text{min}}=50$ Hz, $f_{\text{max}}=14$ kHz.
- **Config B.** $n_{\text{mel}}=64$, $\text{hop} = 160$, $f_{\text{min}}=20$ Hz, $f_{\text{max}}=16$ kHz.

3. Methods

3.1. Backbones

We use three CNN backbones from the `timm` library [11], each initialized from ImageNet-21K pretrained weights [12] (Table 1). Each backbone is paired with a Sound Event Detection head. A 1×1 convolution produces frame-level logits over 206 classes, and an attention module produces frame weights that are normalized across time and applied to the frame-level logits to yield clip-level predictions.

Table 1: Backbones in the final ensemble. Each backbone is trained under both mel configurations, giving six models in total. IN-21K denotes ImageNet-21K and NS denotes NoisyStudent.

Backbone (timm name)	Pretrain	Params
<code>tf_efficientnetv2_s_in21k</code>	IN-21K	21 M
<code>eca_nfnet_l0</code>	IN-21K	24 M
<code>tf_efficientnet_b0_ns</code>	IN-21K + NS	5 M

Cross-fold weight averaging (uniform model soup). For each backbone and configuration pair we train five cross-validation folds and arithmetic-mean their final-epoch weight tensors element-wise into a single model before export [10]. This weight averaging smooths fold-specific noise and gave a small but consistent OOF improvement. Its largest practical effect was halving export size and inference latency relative to a per-fold ensemble.

3.2. Training

Stage 1: Historical pretraining. We pretrain each backbone on the BirdCLEF 2025 official training audio mapped onto the 2026 species set, using focal-BCE [7] with $\gamma=2.0$. The 2025 stage uses a single 90/10 train/val split of the 2025 audio, stratified by species (singleton species, those with only one clip, are kept entirely in train). No cross-validation is performed at this stage. There are no Pantanal soundscapes and no soundscape augmentation at this stage. The optimizer is AdamW with $\eta_{\text{head}}=3 \times 10^{-3}$ and $\eta_{\text{backbone}}=3 \times 10^{-4}$ on a cosine schedule with two warmup epochs, capped at 20 epochs. EfficientNetV2-S and `eca_nfnet_l0` ran the full 20 epochs, reaching val ROC-AUC 0.963 and 0.969 on the held-out 2025 split. EfficientNet-B0 early-stopped at epoch 13 (val ROC-AUC 0.948). All five stage 2 folds inherit the same single stage 1 checkpoint. When stage 2 begins we discard the classification head, reinitialise it for the 2026 label space, and build a fresh AdamW optimizer and cosine schedule.

Stage 2: Cross-validation fine-tuning. We fine-tune on the 2026 labeled audio using 5-fold stratified cross-validation, stratified by species frequency. The optimizer is AdamW [13] with $\eta_{\text{head}}=10^{-3}$, $\eta_{\text{backbone}}=10^{-4}$, cosine annealing with linear warmup, batch size 32, 150 epochs, and early stopping with patience 30 epochs. The loss is focal-BCE ($\gamma=2.0$) plus label smoothing 0.005. Augmentations include mixup [8] ($\alpha=0.5$), SpecAugment [9] (two time and two frequency masks), Gaussian noise, and random volume jitter.

Soundscape augmentation. The single most impactful design choice in our system. With probability p , we replace the silent or low-energy background of a training clip with a 5-second window drawn from a Pantanal soundscape recording at signal-to-noise ratio in [6, 20] dB. We sweep $p \in \{0.0, 0.3, 0.5, 0.7\}$ in Section 4.2..

3.3. Soundscape-Validated Ensemble Selection

Standard ensemble-tuning approaches use OOF predictions on held-out folds to optimize per-model weights, typically maximizing AUC via Nelder-Mead or BFGS. We replace the OOF objective with the labeled soundscape distribution itself.

For every souped candidate model:

1. Run the model on each of the 66 labeled Pantanal soundscape files using the same OpenVINO FP16 [14] inference path used at submission time.
2. Aggregate per-window predictions to a 792×234 ma-

trix (66×12 output bins, full 234-class output vector matching the sample-submission column order).

3. Restrict the target matrix to species with at least one positive across the 66 files (68 of the 234 columns survive).
4. Compute macro-averaged ROC-AUC on that restricted target. We call this the model’s *soundscape AUC* (ss-AUC).
5. Run Nelder-Mead over the simplex of weights to maximize the macro-averaged ROC-AUC of the weighted ensemble on the same target.

The output is both a per-model quality score and an ensemble weighting that is, by construction, aligned with the leaderboard metric on the closest available proxy distribution.

3.4. Soundscape-Aware Training

Building on the procedure above, we modified the training loop to compute soundscape AUC at the end of each epoch and to use it as the early-stopping criterion. The implementation pre-loads the 792 labeled windows once at the start of training and runs batched inference at the end of each epoch (about 1 second of overhead per epoch on a single A100). A converged checkpoint scored 0.960 ss-AUC under the trainer-side soundscape evaluator versus 0.967 ss-AUC under the standalone OpenVINO evaluator. The small gap is due to PyTorch and OpenVINO numerics together with BF16 and FP16 precision differences.

3.5. Inference

At submission time, we slide a 5-second window with a 2.5-second hop over each soundscape, producing 24 overlapping windows per file. Every output bin averages all windows whose span overlaps that bin, giving interior bins three contributing predictions. Predictions are aggregated across the ensemble using the soundscape-validated weights from Section 3.3. All models are exported through ONNX to OpenVINO FP16. A 6-model ensemble runs in roughly 64 minutes for 700 test soundscapes on Kaggle’s 4-core CPU, and a 3-model ensemble runs in roughly 42 minutes. Both fit within the 90-minute submission budget.

4. Results

We evaluate every system using the same three metrics. **OOF AUC** is macro-averaged ROC-AUC on held-out cross-validation folds drawn from the Xeno-Canto training pool. **ss-AUC** is macro-averaged ROC-AUC on the 66 labeled `train_soundscapes` files, restricted to the 68 species that have at least one positive label there. **LB AUC** is the public-leaderboard macro-averaged ROC-AUC on the hidden 700 test soundscapes, scored automatically by Kaggle.

4.1. Quantitative Performance

Table 2: Public-leaderboard progression. “Selection” indicates the metric used to choose ensemble weights. “n/a” marks systems that predate the soundscape-validated procedure.

Configuration	Selection	ss-AUC	LB AUC
Baseline (no soundscape aug)	OOF	n/a	0.705
+ Soundscape aug $p = 0.5$	OOF	n/a	0.823
+ Historical pretrain	OOF	n/a	0.825
+ Soundscape r1 added (v39)	OOF	n/a	0.830
Pseudo-labels round 5	OOF	n/a	0.747
6-model with <code>sc_v2</code> ($p = 0.7$)	OOF	n/a	0.822
Mega ensemble (12 models)	OOF	n/a	0.811
3-model cfg A, ss-validated (v47)	ss-AUC	0.974	0.829
6-model ss-optimized (v48)	ss-AUC	0.983	0.828
<i>With soundscape-aware training:</i>			
6-model <code>ssval_aug05</code> (v49)	ss-AUC	0.985	0.833

Table 2 summarises our progression on the public leaderboard. The largest single jump is from soundscape augmentation, moving LB AUC from 0.705 to 0.823 (+0.118). Our best submitted result is a 6-model `ssval_aug05` ensemble at LB AUC **0.833**, trained with soundscape-aware early stopping and weighted via soundscape-validated selection. The combination of training-time and selection-time use of the labeled soundscapes adds +0.003 LB over the OOF-tuned 0.830 baseline, and +0.005 LB over the soundscape-tuned-only 0.828 baseline.

4.2. Ablation: Soundscape Augmentation Probability

We swept the soundscape-augmentation probability $p \in \{0.0, 0.3, 0.5, 0.7\}$. OOF AUC stayed essentially flat across the sweep, in the range 0.940 to 0.943, while LB AUC moved from 0.705 (at $p=0$, no soundscape data) to 0.816 (at $p=0.3$), 0.825 (at $p=0.5$, the best setting), and 0.817 (at $p=0.7$, where the foreground is too easily dominated by background). OOF AUC is insensitive to the design choice that has the largest effect on LB AUC.

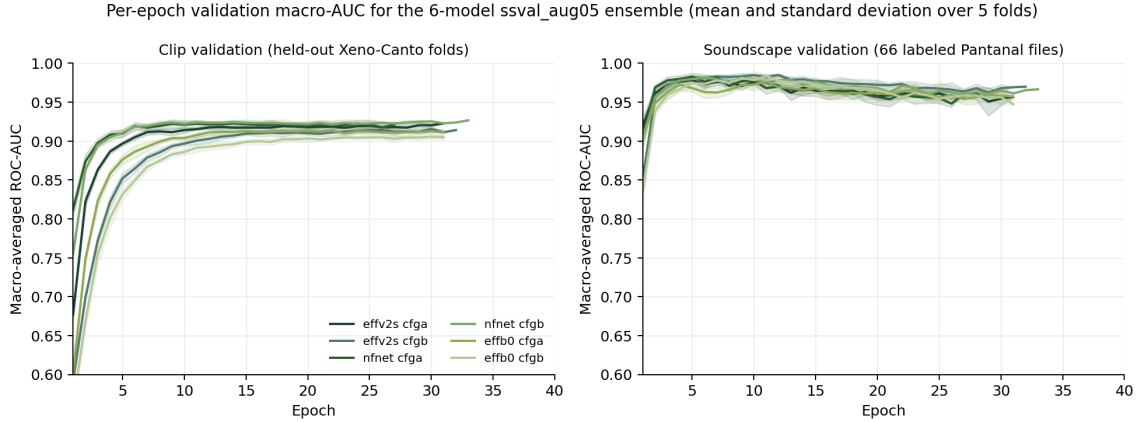


Figure 1: Per-epoch validation macro-AUC for the six souped models in the **ssval_aug05** ensemble, averaged over the five cross-validation folds (shaded band shows standard deviation). Left: clip validation AUC on the held-out Xeno-Canto folds. Right: soundscape AUC on the 66 labeled **train_soundscape**s files. Both metrics rise quickly in the first 5 epochs and plateau by epoch 20, but the clip metric saturates near 0.94 while the soundscape metric reaches roughly 0.99 because the labeled soundscapes also appear in the training pool (Section 2.2.).

4.3. The OOF and LB Anti-Correlation

Table 3: Per-model OOF AUC versus ss-AUC. The top group is the production **combined_sc_aug05** family. The bottom group lists auxiliary candidates (**soundscape_r1**, ConvNeXt, PANNs) that the OOF tuner upweighted but that score below 0.65 ss-AUC on the labeled soundscape set.

Model	OOF AUC	ss-AUC	Verdict
effv2s_cfga	0.942	0.967	ship
nfnet_cfgb	0.942	0.966	ship
nfnet_cfga	0.942	0.966	ship
effb0_cfga	0.942	0.962	ship
effb0_cfgb	0.942	0.945	ship
effv2s_cfgb	0.942	0.916	ship
sc_r1/effv2s_cfga	0.951	0.627	drop
sc_r1/nfnet_cfga	0.951	0.595	drop
sc_r1/effb0_cfga	0.951	0.580	drop
convnext_r2/cfga	0.912	0.567	drop
panns_r1/cnn14	0.905	0.553	drop

Table 3 reports the central observation in our experiments. The model with the highest OOF AUC, **soundscape_r1/effv2s_cfga** at 0.951, achieves only 0.627 ss-AUC on the labeled soundscapes. The same OOF-tuned ensemble assigned 0.41 weight to that model in our prior submissions, while assigning small or zero weight to **combined_sc_aug05/effv2s_cfga**, a model with lower OOF AUC (0.942) but a 0.34 absolute higher ss-AUC of 0.967.

A direct comparison of the same six models, weighted once by OOF AUC and once by ss-AUC, gave LB scores of 0.811 and 0.829 respectively, an improvement of +0.018 LB attributable solely to the choice of selection metric.

4.4. Negative Result: Pseudo-Labeling

Table 4: Pseudo-labelling improves OOF and reduces LB. The OOF AUC column is on held-out training folds, and LB AUC is the public leaderboard. The round-7 retrain used the soundscape-aware-trained teacher and we report ss-AUC for it instead of OOF AUC.

System	OOF AUC	LB AUC	Δ LB
Best non-pseudo ensemble	0.942	0.825	n/a
+ Pseudo round 4	0.945	0.737	-0.088
+ Pseudo round 5	0.949	0.747	-0.078
+ Pseudo round 7 (v49 teacher)	0.989 (ss)	0.830	-0.003

We generated pseudo labels by running the then-best ensemble on the roughly 10,500 unlabeled Pantanal soundscapes, kept rows where any class probability cleared a sample-count-aware threshold (0.15 for rare classes, 0.20 medium, 0.25 common), and retrained on the combined dataset (Table 4). Two early iterations improved OOF AUC by up to +0.007 but degraded LB AUC by 0.078 to 0.088. A third round (pseudo round 7) used the LB-best **ssval_aug05** ensemble as teacher and produced a hybrid ensemble at ss-AUC 0.986, yet still came in below the v49 baseline on the leaderboard (LB AUC 0.830 versus 0.833). Pseudo labels generated on out-of-distribution audio carry the teacher’s systematic errors. Training on those errors as if they were ground truth reinforces rather than corrects them. We do not use pseudo labels in the final system.

4.5. Negative Result: Per-Class Ensemble Weights

Macro-averaged ROC-AUC decomposes into an average of independent per-class ROC-AUCs, so the joint ensemble-weight optimization in Section 3.3. can be split into one Nelder-Mead subproblem per active class. The

unconstrained per-class fit improved the selection-set ss-AUC from 0.985 to 0.988, a gain of +0.0024. To check whether the gain generalises we ran leave-half-out cross-validation across the 66 files. We fit weights on 33 files, score on the held-out 33, and average over both halves. The held-out gain over the global solution was +0.0001 on the 6-model pool and +0.0005 on a 12-model pool. Both are below the split-half disagreement of about 0.005 ss-AUC. An ℓ_2 -shrinkage sweep toward the global solution did not recover any held-out lift either. We conclude that on a 66-file selection set the marginal optimisation budget for per-class weights is exhausted, and we did not push a per-class kernel.

4.6. Qualitative Per-Species Performance

Figure 2 disaggregates the global ss-AUC into a per-species distribution over the 68 species that have at least one positive in the labeled soundscapes. The lower decile sits above 0.93, with no class falling below 0.87 ss-AUC.

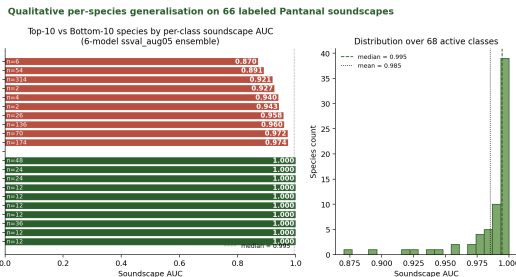


Figure 2: Per-species ss-AUC for the 6-model `ssval_aug05` ensemble on the 66 labeled `train_soundscapes` files. Left: the 10 best- and 10 worst-recovered species by per-class ROC-AUC, with bar labels showing window counts. Right: histogram over all 68 active species. The median ss-AUC is 0.995. Note that the labeled soundscapes are also in the training set, so this distribution is an upper bound on training-set recall rather than an out-of-distribution generalisation estimate.

5. Conclusion and Future Work

We presented a competitive system for BirdCLEF+ 2026 that reaches a public-leaderboard macro-averaged ROC-AUC of 0.833. The system combines historical pretraining, soundscape-domain augmentation, cross-fold weight averaging, soundscape-validated ensemble selection, soundscape-aware early stopping, and overlapping-window OpenVINO inference. The contribution of this work is independent of the specific backbone. Validating ensemble weights against the labeled deployment distribution rather than held-out training-distribution folds yields an immediate and robust LB improvement, and reveals model-quality differences that OOF AUC obscures. The same framework explains why naive pseudo-labelling consistently degrades LB despite improving OOF.

Limitation. The 66 labeled soundscapes serve simultaneously as training data, the early-stopping signal, and the ensemble-selection target. Our reported ss-AUC of

0.985 is therefore an upper bound on training-set recall rather than a held-out estimate. The leave-half-out CV in Section 4.5. puts the same number at about 0.980, and the gap to LB AUC 0.833 shows that training-set recall is not a reliable proxy for the true deployment distribution.

Future Work. Closing the residual gap likely requires more LB-aligned signal. We plan to hold out a small fraction of the labeled soundscapes from training (for example, 10 of the 66) to recover an honest validation set at the cost of fewer in-domain examples. We also plan to evaluate per-species threshold calibration and post-hoc temperature scaling on output probabilities [15].

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